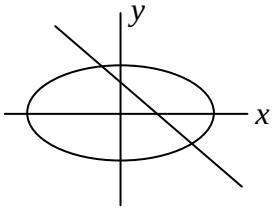
MATHEMATICS

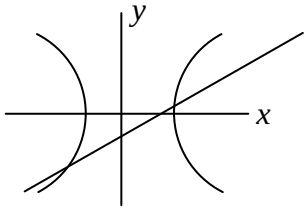
SECTION – I
Single Correct Choice Type

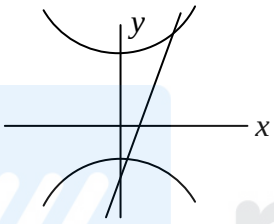
This section contains 4 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

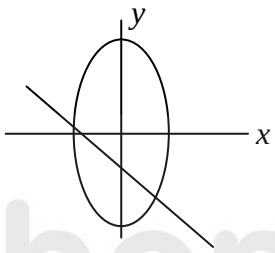
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1. Let $a, b \in R$ where $ab \neq 0$, then the graph of the straight line $ax - y + b = 0$ and the conic section $bx^2 + ay^2 = ab$ can be

(a) 

(b) 

(c) 

(d) 

2. $f : R \to R,\ f(x) = \dfrac{x^2 + bx + 1}{x^2 + 2x + b}$, if function $f(x)$ and $\dfrac{1}{f(x)}$ have same bounded set as its range, then value of b is

(a) $2\sqrt{3} - 2$ (b) $2\sqrt{3} + 2$ (c) $2\sqrt{2} - 2$ (d) $2\sqrt{2} + 2$

3. If the differential equation of a curve, passing through $\left(0, -\dfrac{\pi}{4}\right)$ and $(t, 0)$ is $\cos y\left(\dfrac{dy}{dx} + e^{-x}\right) + \sin y\left(e^{-x} - \dfrac{dy}{dx}\right) = e^{e^{-x}}$, then the value of $t.e^{e^{-t}}$ is

(a) -1 (b) 1 (c) 2 (d) -2

4. The number of positive integral solution of the equation $\begin{vmatrix} y^3+1 & y^2z & y^2x \\ yz^2 & z^3+1 & z^2x \\ yx^2 & x^2z & x^3+1 \end{vmatrix} = 11$ is

(a) 1 (b) 2 (c) 3 (d) 4

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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5. Let z_1, z_2, z_3 be the coordinate of vertices A, B, C of the triangle if $W_1 = z_1 - z_2$, $W_2 = z_3 - z_2$ and $\angle A = 90°$, then

(a) $\text{Re}\left(\frac{W_2}{W_1}\right) = 1$ (b) $\text{Re}(W_1\overline{W_2}) = |W_1|^2$

(c) $\text{Re}(W_1\overline{W_2}) = |W_2|^2$ (d) $\text{Re}\left(\frac{W_1}{W_2}\right) = 1$

6. Let $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 ..... + a_{2n}x^{2n}$, where n is odd integer, if

$S_1 = a_0 + a_4 + a_8 + ...$

$S_2 = a_1 + a_5 + a_9 + ...$

$S_3 = a_2 + a_6 + a_{10} + ...$

$S_4 = a_3 + a_7 + a_{11} + ...$

then

(a) $S_1 = S_3$ (b) $S_2 = S_4$

(c) $S_2 + S_4 = 0$ (d) either $S_1 = S_2 = S_3$ or $S_1 = S_3 = S_4$

7. $\cos\alpha$ is a root of the equation $169x^2 - 26x - 35 = 0$, $-1 < x < 0$, then $\sin 2\alpha$ is

(a) $\frac{144}{169}$ (b) $-\frac{144}{169}$ (c) $\frac{120}{169}$ (d) $-\frac{120}{169}$

8. A solution (x, y) of the system of equations $x - y = \frac{1}{3}$ and $\cos^2(\pi x) - \sin^2(\pi y) = \frac{1}{2}$ is given by

(a) $\left(\frac{7}{6}, \frac{5}{6}\right)$ (b) $\left(\frac{2}{3}, \frac{1}{3}\right)$ (c) $\left(-\frac{5}{6}, -\frac{7}{6}\right)$ (d) $\left(\frac{13}{6}, \frac{11}{6}\right)$

SECTION – III
Reasoning Type

This section contains 4 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

9. **Statement-1 :** For a conic section if its two foci are (2, 3), (10, 12) and its directrix is $8x + 9y - 50 = 0$, then it must be hyperbola.

Statement-2 : For a hyperbola, line joining the foci is perpendicular to directrix.

(a) A (b) B (c) C (d) D

10. **Statement-1 :** $\lim_{x\to 0}\left(\left(\frac{1+2x}{1+3x}\right)^{\frac{1}{x^2}} e^{\frac{1}{x}}\right) = \lim_{x\to 0} e^{\left\{\frac{1}{x^2}\left(\frac{1+2x}{1+3x}-1\right)+\frac{1}{x}\right\}} = e^{\lim_{x\to 0}\frac{1}{(1+3x)x}+\frac{1}{x}} = e^{\lim_{x\to 0}\frac{3}{1+3x}} = e^3$.

Statement-2 : If $\lim_{x\to a} f(x) = l_1$ and $\lim_{x\to a} g(x) = l_2$, then $\lim_{x\to a}(f(x)g(x)) = l_1.l_2$

(where l_1 and l_2 are finite).

(a) A (b) B (c) C (d) D

11. **Statement 1 :** $\lim\limits_{t\to 0}\int_0^{2\pi}\frac{|\sin(x+t)-\sin x|}{|t|}dx$ is equals to 4.

Statement 2 : $f'(x)=\lim\limits_{h\to 0}\frac{f(x+h)-f(x)}{h}$.

(a) A (b) B (c) C (d) D

12. **Statement-1 :** Let $f:R\to R,\ f(x)$ be a integrable function then

$$\int_0^1 f^2(x)\,dx \geq \left(\int_0^1 f(x)\,dx\right)^2.$$

Statement-2 : $\sum\limits_{r=1}^{n}\left(\frac{a_r^2}{n}\right)\geq\left(\sum\limits_{r=1}^{n}\frac{a_r}{n}\right)^2$.

(a) A (b) B (c) C (d) D

SECTION–IV
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

Consider an isosceles triangle *ABC* with base *BC* = 8 units and $\angle B = \angle C$. A line cuts the base externally at *D* such that it make angle θ with base and bisects the side which is nearer to *D*.

13. If $\angle B = 45°$, then the area of quadrilateral which the straight line cuts from the given triangle is

(a) $\frac{3+\tan\theta}{1+\tan\theta}$ (b) $\frac{3+5\tan\theta}{1+\tan\theta}$ (c) $\frac{2(3+5\tan\theta)}{1+\tan\theta}$ (d) $\frac{4(3+5\tan\theta)}{1+\tan\theta}$

14. If $\angle B = 45°$, then length of portion of straight line inside the triangle may lie in the range

(a) (2, 4) (b) $(2\sqrt{2}, 4)$ (c) $(\sqrt{2}, 2)$ (d) none of these

15. If the vertices of triangle *ABC* lie on the curve $x^3 + y^3 + 3xy = 1$, then number of such possible triangles is

(a) 1 (b) 2 (c) 3 (d) 4

Passage-II

$f(x) = a_0 + a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + \ldots a_n \cos nx$

$g(x) = b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + \ldots b_n \sin nx$

16. If $b_r = \dfrac{1}{r+1}$, then $\lim\limits_{n\to\infty} \int_0^{\pi} g(x)\,dx$ is

(a) $\frac{1}{2}\ln\frac{1}{2}$ (b) $\frac{1}{2}\ln 2$ (c) $\ln 2$ (d) $2\ln 2$

17. $\sum\limits_{k=0}^{n} \int_0^{2\pi} f(x).(\cos kx)\,dx$ is

(a) $\pi f(0)$ (b) $\pi f(1)$ (c) $\pi f(-1)$ (d) 0

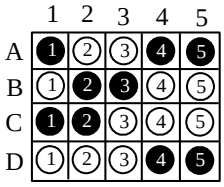
18. $\sum\limits_{k=0}^{n} \int_0^{2\pi} g(x).(\cos kx)\,dx$ is

(a) $\pi f(0)$ (b) $\pi g(1)$ (c) $\pi g(-1)$ (d) 0

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

Column I (Function)		Column II (Property)
(A)	$f: N \to I \quad f(x) = (x+1)^2 + x - [\sqrt{(x+1)^2 + (x+1)}]^2$, then $f(x)$ will be (where [.] is greatest integer function)	**1.** one-one function
(B)	Let $f(x) = \frac{\log x}{x}$, and solutions of $f(x) = k$ be denoted by $g(k)$ {where $k < 0$}, $g: (-\infty, 0) \to (0, 1)$, then $y = g(x)$ will be	**2.** many one function
(C)	Let $f: R \to R$ be a continuous function such that $\lvert f(x) - f(y) \rvert \geq \lvert x - y \rvert$ for all $x, y \in R$, then $f(x)$ will be	**3.** onto function
(D)	$f: R \sim \{0\} \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ such that $f(x) = \int_0^x \frac{1}{1+t^2} dt + \int_0^{1/x} \frac{1}{1+t^2} dt$ ($x \neq 0$), now $f(x)$ will be	**4.** into function **5.** neither one-one nor onto

2.

Column I	**Column II**
(A) Let $\lvert z_1\rvert=\lvert z_2\rvert=\lvert z_3\rvert=1$, $\frac{z_1^2}{z_2z_3}+\frac{z_2^2}{z_1z_3}+\frac{z_3^2}{z_1z_2}+1=0$ and $\lvert z_1+z_2+z_3\rvert=k$, then k can be	**1.** 1
(B) Let $\lvert z\rvert=1$ and $a\le\lvert 1-z\rvert+\lvert 1+z^2\rvert\le b$, and k be a integer such that $k\in[a\ b]$, then k can be	**2.** 2
(C) Let $\lvert z\rvert=1,\lvert 1-z\rvert=k\sin\frac{\theta}{2}$, (where $\theta=\arg(z)$), then (k+2) can be	**3.** 3
(D) Let $f(x)=x^4+ax^3+bx^2+cx+d$ (where a, b, c, d are real coefficients) and $f(x)=0$ has real roots. If $\lvert f(i)\rvert=1$ and $a=b=c=d=k$, then k can be	**4.** 4 **5.** 0

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